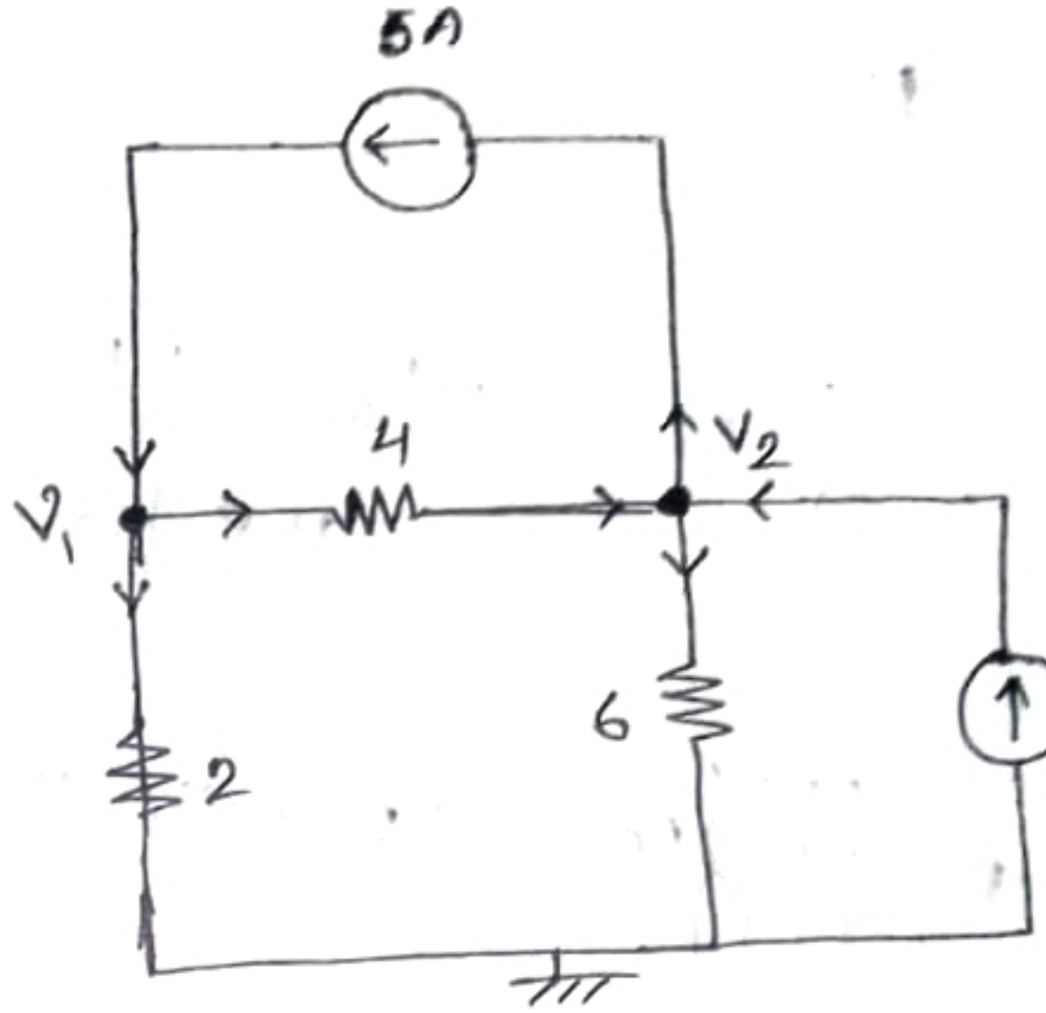
01. (b)



Soln: Applying KCL at node V_1.

$$-5 + \frac{V_1 - 0}{2} + \frac{V_1 - V_2}{4} = 0$$

$$\text{or } \frac{-20 + 2V_1 + V_1 - V_2}{4} = 0$$

$$\text{or, } 3V_1 - V_2 = 20 \quad \text{---(i)}$$

Applying KCL at node V_2

$$-10 + 5 + \frac{V_2 - 0}{6} - \frac{V_1 - V_2}{4} = 0$$

$$\text{or, } \frac{-120 + 60 + 2V_2 - 3V_1 + 3V_2}{12} = 0$$

$$\text{or, } 3V_1 - 5V_2 = -60 \quad \text{---(ii)}$$

Solve the eqn's using calculator

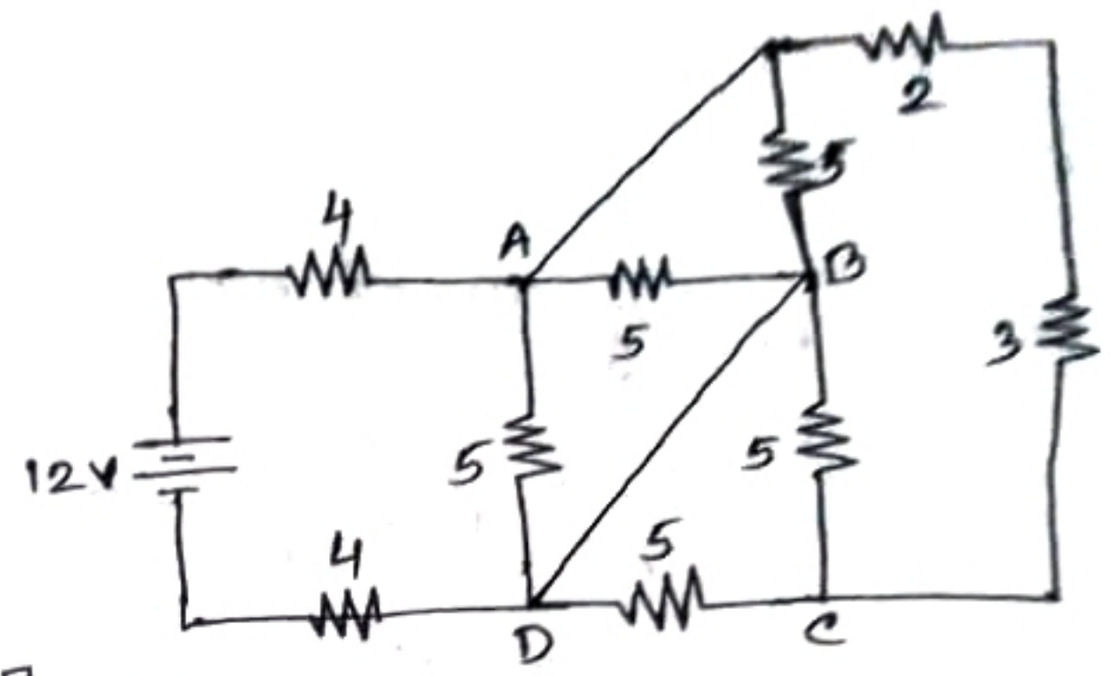
$$V_1 = 13.3\ V$$

$$V_2 = 20\ V$$

1(c)

Soln:

$$R_{eq} = [[[\{(R_8 \,||\, R_9) + R_6 + R_7\} \,||\, R_5] \,||\, R_4] \,||\, R_3] + R_2 + R_1$$



$$R_{eq} = [[[\{(5 \,||\, 5) + 2 + 3\} \,||\, 5] \,||\, 5] \,||\, 5] + 4 + 4$$

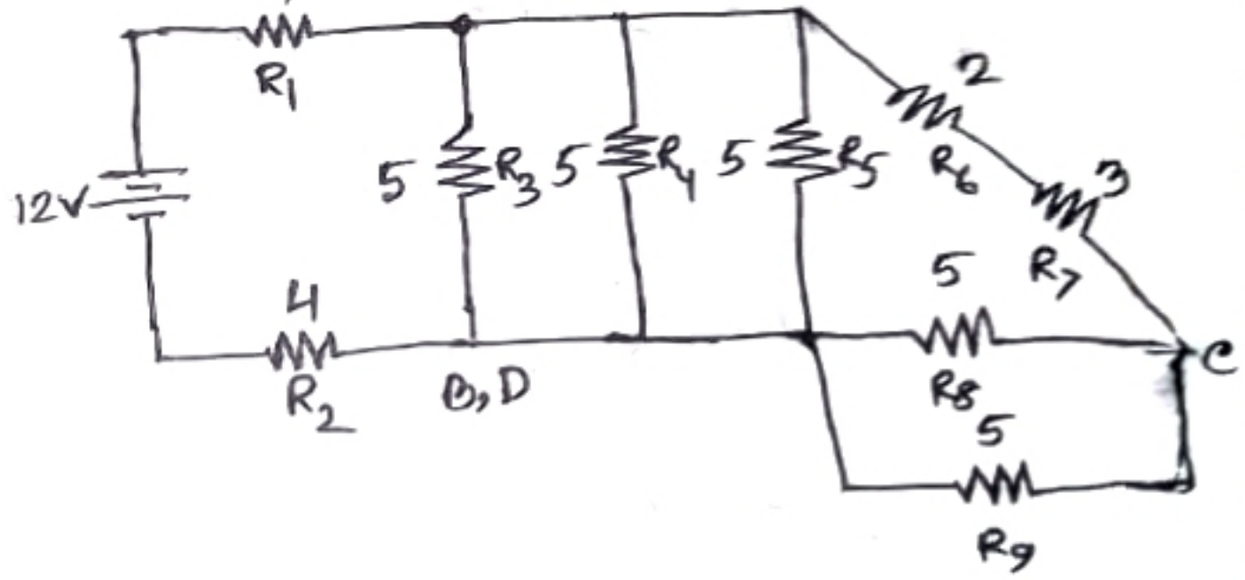
$$= [[[\{(2.5 + 5) \,||\, 5\} \,||\, 5] \,||\, 5] + 8$$

$$= [[(7.5 \,||\, 5) \,||\, 5] \,||\, 5] + 8$$

$$= \{(3 \,||\, 5) \,||\, 5\} + 8$$

$$= (1.9 \,||\, 5) + 8$$

$$= 1.4 + 8$$

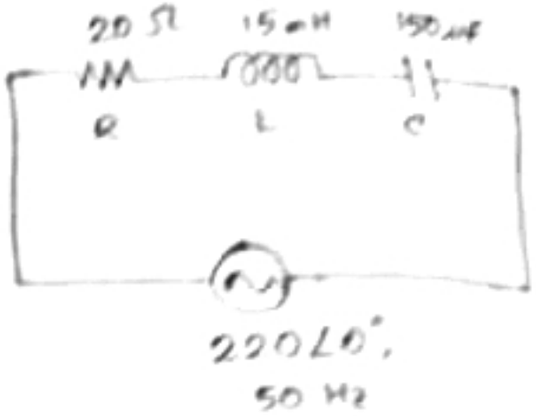


$$\therefore R_{eq} = 9.4\ \Omega$$

$$\therefore I_{return} = \frac{V}{R_{eq}} = \frac{12}{9.4} = 1.3\ A$$

$$\therefore I_{return} = 1.3\ A$$

3(c)



Soln. Here, $X_L = 2\pi fL$

$= 2 \times \pi \times 50 \times 15 \times 10^{-3}$

$= 4.71\ \Omega$

and, $X_c = \frac{1}{2\pi fC}$

$= \frac{1}{2 \times \pi \times 50 \times 150 \times 10^{-6}}$

$= 21.22\ \Omega$

$\therefore$ Impedance, $Z = R + j(X_L - X_c)$

$= 20 + j(4.71 - 21.22)$

$= 20 - j16.51$

$\therefore Z = 25.93 \angle 39.54^\circ\ \Omega$

Total current,

$I_T = \frac{V}{Z} = \frac{220\angle 0^\circ}{25.93\angle -39.54^\circ}$

$\therefore I_T = 8.48 \angle 39.54^\circ\ A$

$\therefore V_R = I_T \times R = (8.48\angle 39.54^\circ) \times 20$

$\therefore V_R = 169.6 \angle 39.54^\circ\ V$

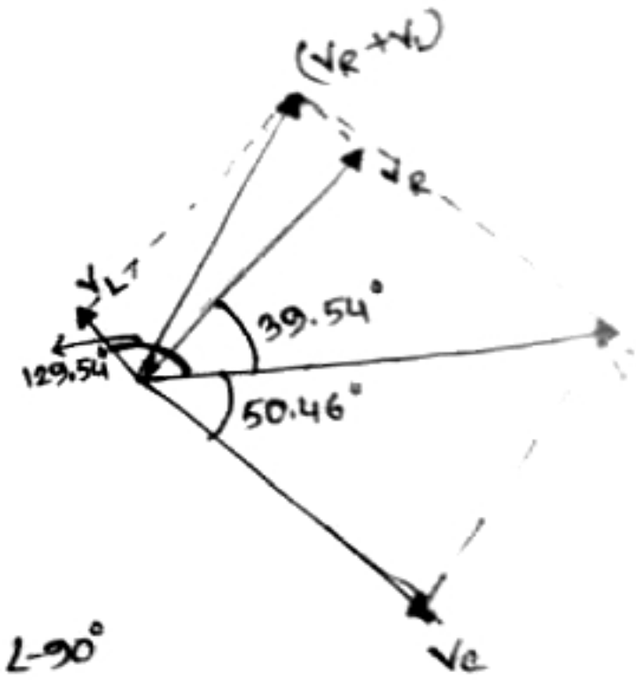
$V_L = I_T \times X_L = (8.48\angle 39.54^\circ) \times 4.71 \angle 90^\circ$

$\therefore V_L = 39.94 \angle 129.54^\circ\ V$

$V_c = I_T \times X_c = (4.84\angle 39.54) \times 21.22 \angle -90^\circ$

$\therefore V_c = 179.95 \angle -50.46^\circ\ V$

Voltage Vector



N.B → Draw the voltage vector by protector

Power absorbed by Resistor, $P_R = (8.48)^2 \times 20$

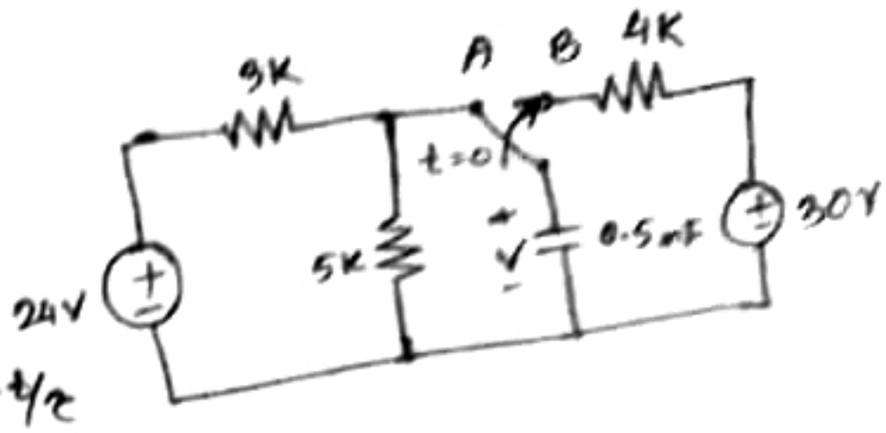
$\therefore P_R = 1438.2\ W$

Power factor $= \cos\theta = \cos(39.54^\circ)$

$\therefore$ P.f $= 0.77$

4(c)

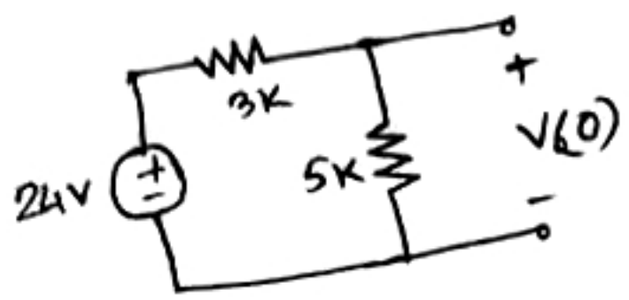
Soln: For sourced circuit,



We have to find,

$$v(t) = v(\infty) + [v(0) - v(\infty)]\, e^{-t/\tau} \quad \text{---} (1)$$

For $v(0)$; $t < 0$.

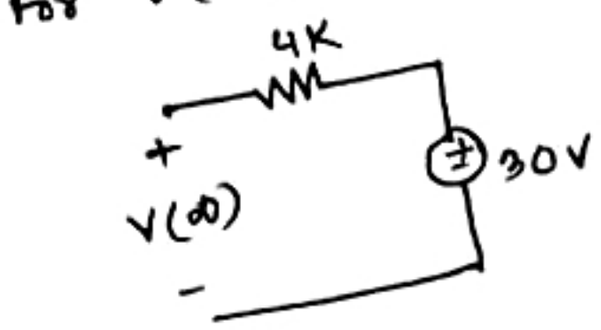


$$\therefore v(0) = \frac{5}{3+5} \times 24$$

$$= \frac{5}{8} \times 24$$

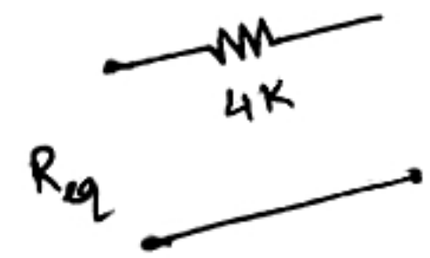
$$\therefore v(0) = 15\ V$$

For $v(\infty)$; $t > 0$.



$$\therefore v(\infty) = 30\ V$$

For τ; $t > 0$



$$\therefore R_{eq} = 4K$$

$$\therefore \tau = R_{eq} C$$

$$= 4 \times 10^{3} \times 0.5 \times 10^{-3}$$

$$= 2\ sec.$$

From equ^{n} (1), we get,

$$v(t) = v(\infty) + [v(0) - v(\infty)]\, e^{-t/\tau}$$

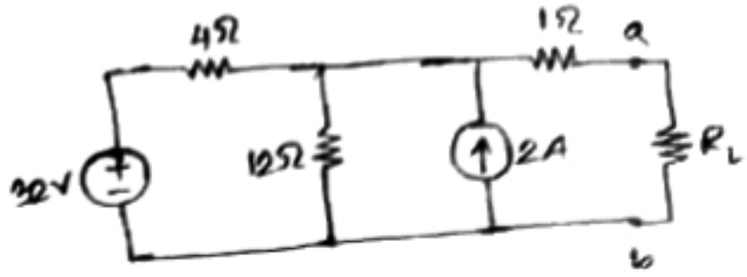
$$\text{or, } 25 = 30 + (15 - 30)\, e^{-t/2}$$

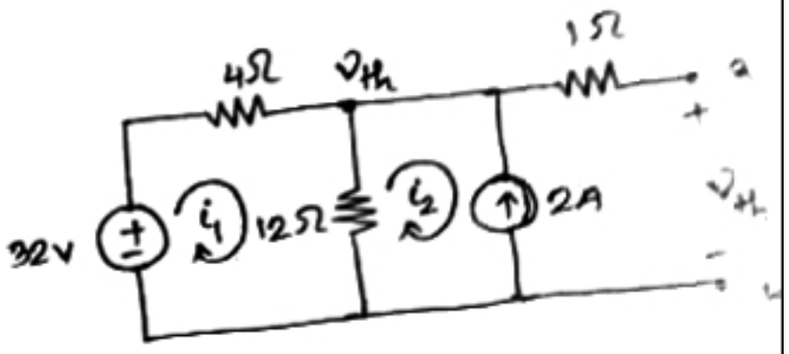
$$\text{or, } 15 e^{-t/2} = 5$$

$$\text{or, } e^{-t/2} = \frac{1}{3}$$

$$\therefore t = 2.197\ sec$$

2 (c)





Sol^n: Applying KVL at loop 1,

$$-32 + 4i_1 + 12(i_1 - i_2) = 0$$

$$\text{or, } 4i_1 + 12i_1 - 12i_2 = 32$$

$$\text{or, } 16i_1 - 12i_2 = 32 \quad \text{---} (1)$$

For loop 2,

$$i_2 = -2\ A$$

From eq^n (1)

$$16i_1 - 12\times(-2) = 32$$

$$\text{or, } 16i_1 + 24 = 32$$

$$\therefore i_1 = 0.5\ A$$

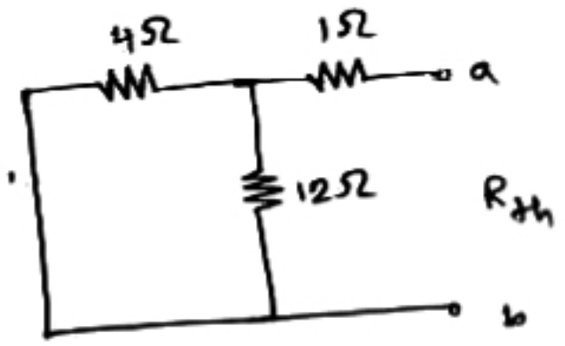
Now, $i_{12} = i_1 - i_2 = 0.5 - (-2) = 0.5 + 2 = 2.5\ A$

$$\therefore V_{th} = 2.5 \times 12 = 30\ V$$

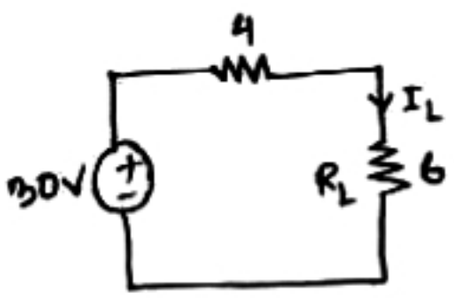
$$R_{th} = (12 \| 4) + 1$$

$$= 3 + 1$$

$$= 4\ \Omega$$



The Thevenin equivalent ckt.



$$\therefore I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{30}{4+6} = \frac{30}{10}$$

$$\therefore I_L = 3A$$